Fundamentals Physics

Eleventh Edition

Halliday

Chapter 25

Capacitance

25-1 Capacitance (1 of 6)

Learning Objectives

25.01 Sketch a schematic diagram of a circuit with a parallel-plate capacitor, a battery, and an open or closed switch.

25.02 In a circuit with a battery, an open switch, and an uncharged capacitor, explain what happens to the conduction electrons when the switch is closed.

25.03 For a capacitor, apply the relationship between the magnitude of charge *q* on either plate ("the charge on the capacitor"), the potential difference *V* between the plates ("the potential across the capacitor"), and the capacitance C of the capacitor.

25-1 Capacitance (2 of 6)

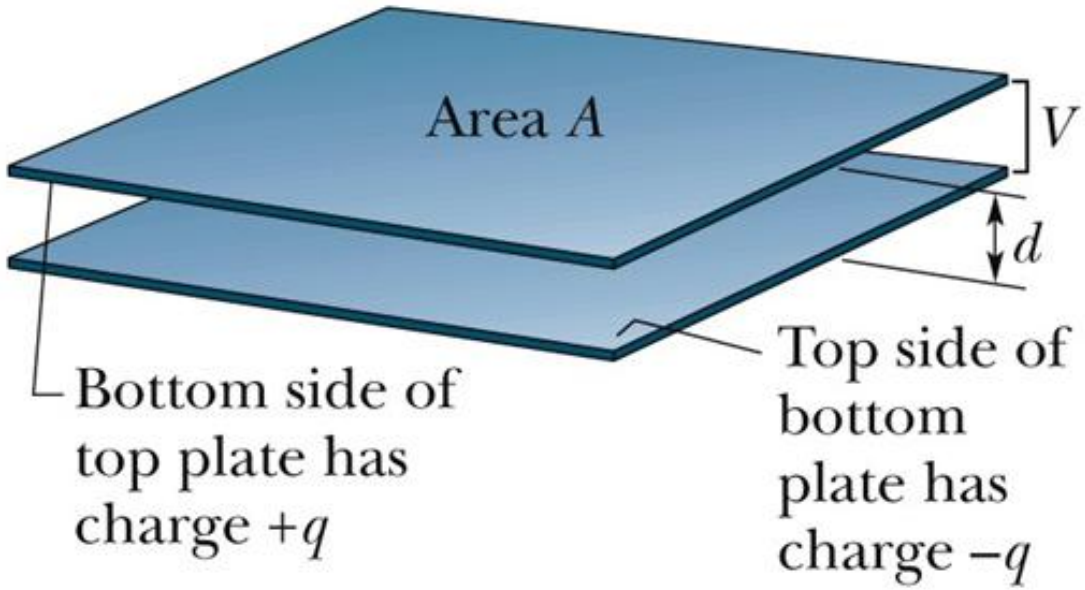
A capacitor consists of two isolated conductors (the plates) with charges $+q$ and $-q$. Its **capacitance** C is defined from

$$q = CV.$$

where V is the potential difference between the plates.

25-1 Capacitance (3 of 6)

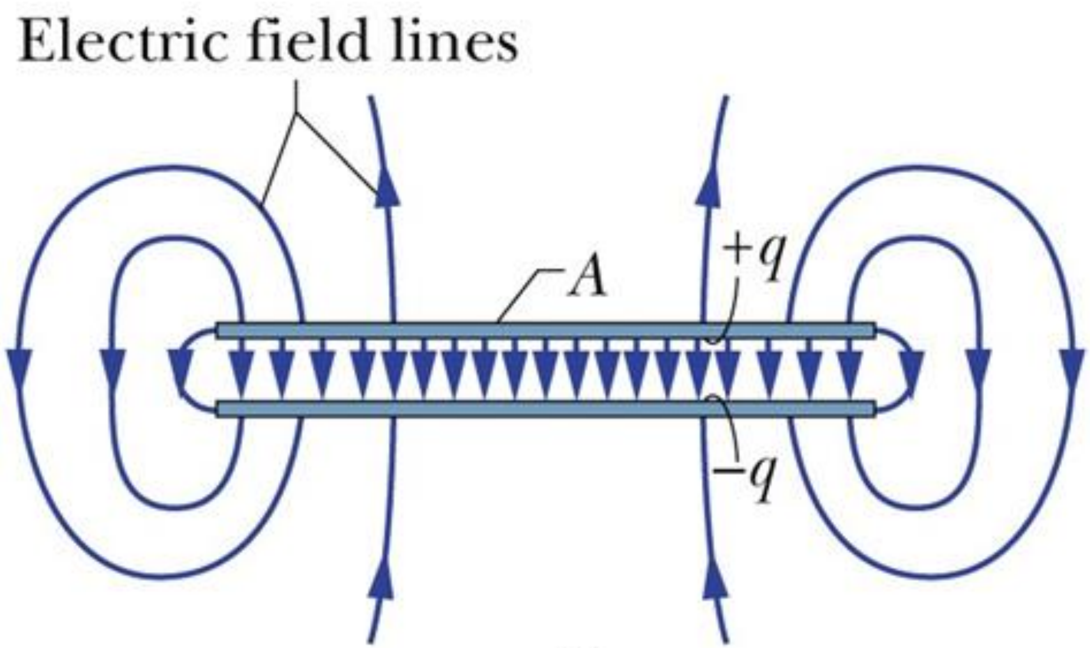
A parallel-plate capacitor, made up of two plates of area *A* separated by a distance *d*. The charges on the facing plate surfaces have the same magnitude *q* but opposite signs



(*a*)

25-1 Capacitance (4 of 6)

As the field lines show, the electric field due to the charged plates is uniform in the central region between the plates. The field is not uniform at the edges of the plates, as indicated by the "fringing" of the field lines there.



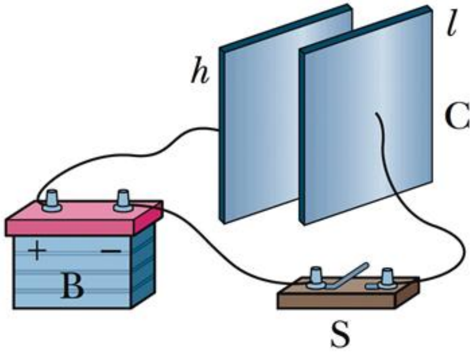
(b)

25-1 Capacitance (5 of 6)

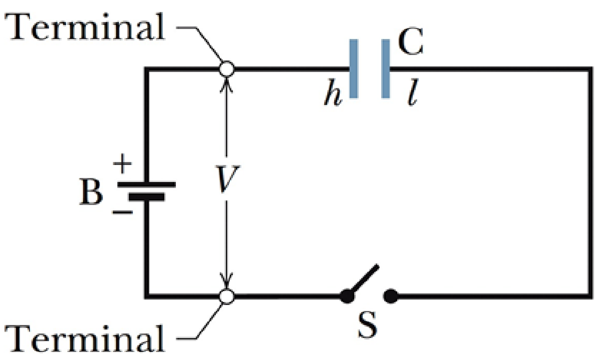
Charging Capacitor

When a circuit with a battery, an open switch, and an uncharged capacitor is completed by closing the switch, conduction electrons shift, leaving the capacitor plates with opposite charges.

25-1 Capacitance (6 of 6)



(*a*)



(*b*)

In Figure. *a*, a battery B, a switch S, an uncharged capacitor C, and interconnecting wires form a circuit. The same circuit is shown in the schematic diagram of Figure. *b*, in which the symbols for a battery, a switch, and a capacitor represent those devices. The battery maintains potential difference V between its terminals. The terminal of higher potential is labeled + and is often called the positive terminal; the terminal of lower potential is labeled − and is often called the negative terminal.

25-2 Calculating the Capacitance (1 of 12)

Learning Objectives

25.04 Explain how Gauss' law is used to find the capacitance of a parallel-plate capacitor.

25.05 For a parallel-plate capacitor, a cylindrical capacitor, a spherical capacitor, and an isolated sphere, calculate the capacitance.

Charging Capacitor

The field drives electrons from capacitor plate *h* to the positive terminal of the battery; thus, plate *h*, losing electrons, becomes positively charged.

The field drives just as many electrons from the negative terminal of the battery to capacitor plate *l*; thus, plate *l*, gaining electrons, becomes negatively charged *just as much* as plate *h*, losing electrons, becomes positively charged.

25-2 Calculating the Capacitance (2 of 12)

Calculating electric field and potential difference

To relate the electric field $\vec{E}$ between the plates of a capacitor to the charge q on either plate, we shall use Gauss' law:

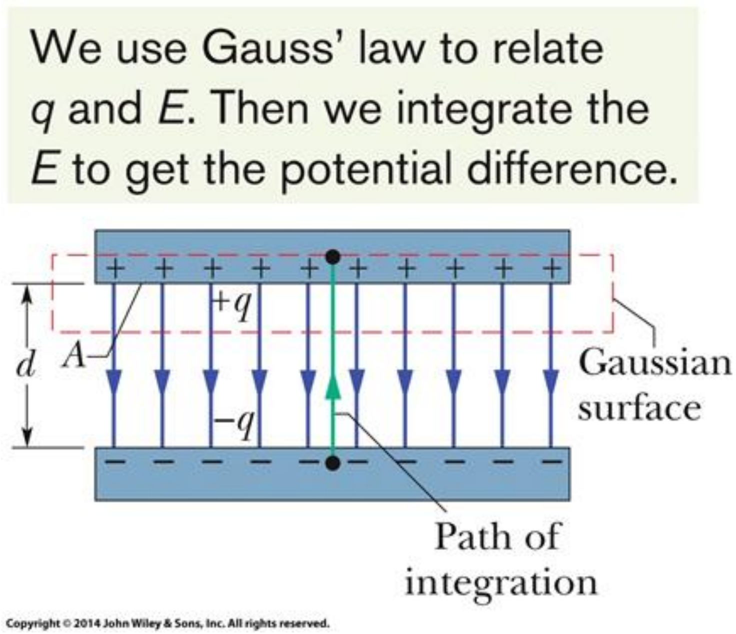
$$\varepsilon_0 \oint \vec{E} \cdot d\vec{A} = q.$$

the potential difference between the plates of a capacitor is related to the field $\vec{E}$ by

$$V_f - V_i = -\int_i^f \vec{E} \cdot d\vec{s},$$

25-2 Calculating the Capacitance (3 of 12)

Letting V represent the difference $V_f = V_i$, we can then recast the above equation as:



$$V = \int_{-}^{+} E\, ds$$

A charged parallel-plate capacitor. A Gaussian surface encloses the charge on the positive plate. The integration is taken along a path extending directly from the negative plate to the positive plate.

25-2 Calculating the Capacitance (4 of 12)

Parallel-Plate Capacitor

We assume, as Figure suggests, that the plates of our parallel-plate capacitor are so large and so close together that we can neglect the fringing of the electric field at the edges of the plates, taking $\vec{E}$ to be constant throughout the region between the plates.

We draw a Gaussian surface that encloses just the charge q on the positive plate

$$q = \varepsilon_0 EA$$

25-2 Calculating the Capacitance (5 of 12)

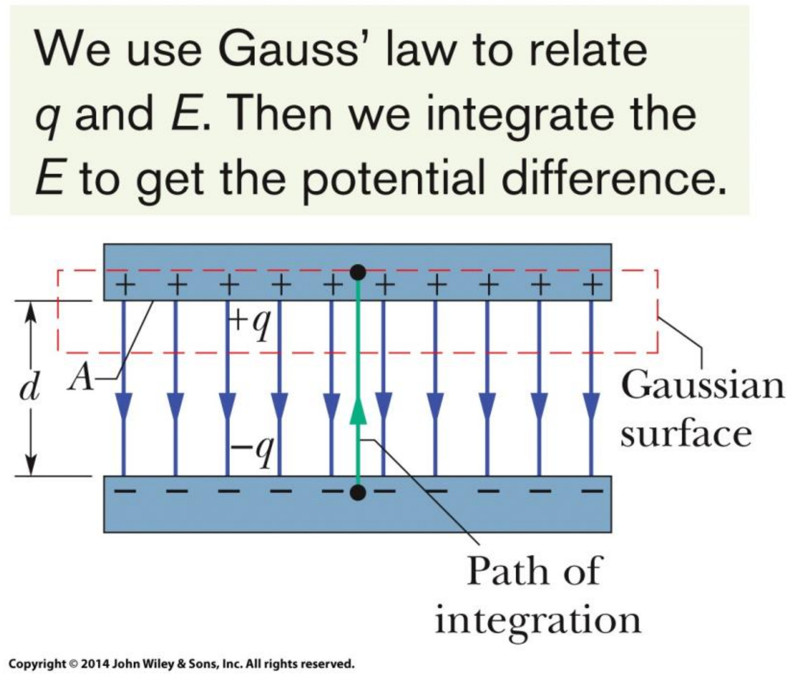
where A is the area of the plate. And therefore,

$$V = \int_{-}^{+} E\, ds = E \int_{0}^{d} ds = Ed.$$

Now if we substitute q in the above relations to $q = CV$, we get,

$$C = \frac{\varepsilon_0 A}{d} \quad (\text{parallel-plate capacitor}).$$

25-2 Calculating the Capacitance (6 of 12)



A charged parallel-plate capacitor. A Gaussian surface encloses the charge on the positive plate. The integration is taken along a path extending directly from the negative plate to the positive plate.

25-2 Calculating the Capacitance (7 of 12)

Cylindrical Capacitor

Figure shows, in cross section, a cylindrical capacitor of length L formed by two coaxial cylinders of radii a and b. We assume that L ? b so that we can neglect the fringing of the electric field that occurs at the ends of the cylinders. Each plate contains a charge of magnitude q. Here, charge and the field magnitude E is related as follows,

$$q = \varepsilon_0 EA = \varepsilon_0 E\left(2\pi rL\right)$$

25-2 Calculating the Capacitance (8 of 12)

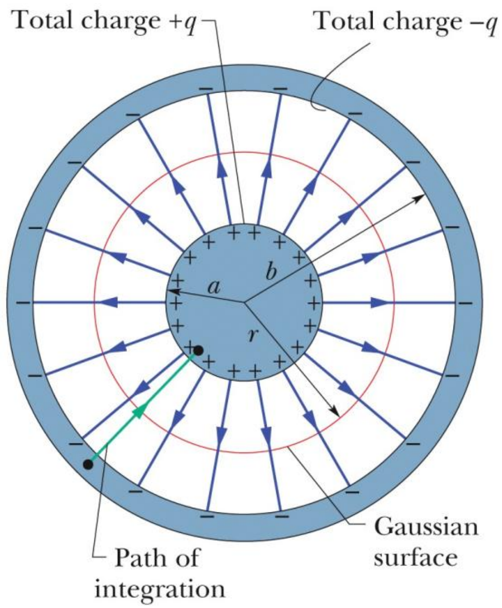
Solving for E field:

$$V = \int_{-}^{+} E\, ds = -\frac{q}{2\pi\varepsilon_0 L}\int_b^a \frac{dr}{r} = \frac{q}{2\pi\varepsilon_0 L}\ln\left(\frac{b}{a}\right)$$

From the relation $C = \frac{q}{V}$, we then have

$$C = 2\pi\varepsilon_0 \frac{L}{\ln\left(\frac{b}{a}\right)} \quad \text{(cylindrical capacitor).}$$

25-2 Calculating the Capacitance (9 of 12)



A cross section of a long cylindrical capacitor, showing a cylindrical Gaussian surface of radius *r* (that encloses the positive plate) and the radial path of integration. This figure also serves to illustrate a spherical capacitor in a cross section through its center.

25-2 Calculating the Capacitance

Others…

For **spherical capacitor** the capacitance is:

$$C = 4\pi\varepsilon_0 \frac{ab}{b-a} \quad (\text{spherical capacitor}).$$

Capacitance of an **isolated sphere**:

$$C = 4\pi\varepsilon_0 R \quad (\text{isolated sphere}).$$

25-2 Calculating the Capacitance (11 of 12)

Checkpoint 2

For capacitors charged by the same battery, does the charge stored by the capacitor increase, decrease, or remain the same in each of the following situations? (a) The plate separation of a parallel-plate capacitor is increased. (b) The radius of the inner cylinder of a cylindrical capacitor is increased. (c) The radius of the outer spherical shell of a spherical capacitor is increased.

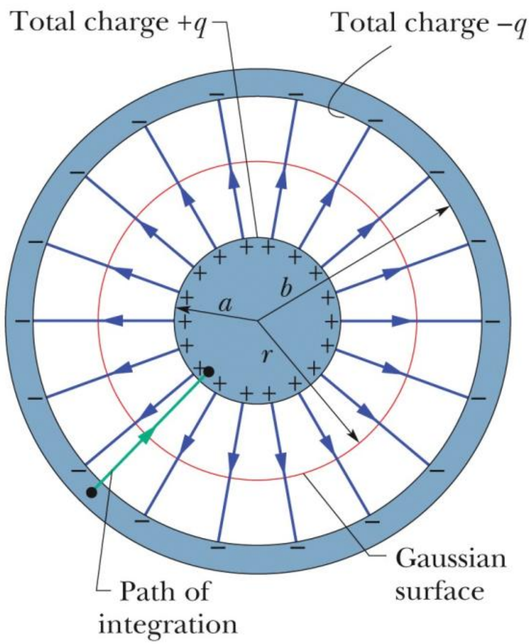
Answer:

(a) decreases

(b) increases

(c) increases

25-2 Calculating the Capacitance (12 of 12)



A cross section of a long cylindrical capacitor, showing a cylindrical Gaussian surface of radius *r* (that encloses the positive plate) and the radial path of integration. This figure also serves to illustrate a spherical capacitor in a cross section through its center.

25-3 Capacitors in Parallel and in Series (1 of 9)

Learning Objectives

25.06 Sketch schematic diagrams for a battery and (a) three capacitors in parallel and (b) three capacitors in series.

25.07 Identify that capacitors in parallel have the same potential difference, which is the same value that their equivalent capacitor has.

25.08 Calculate the equivalent of parallel capacitors.

25.09 Identify that the total charge stored on parallel capacitors the sum of the charges stored on the individual capacitors.

25-3 Capacitors in Parallel and in Series (2 of 9)

25.10 Identify that capacitors in series have the same charge, which is the same value that their equivalent capacitor has.

25.11 Calculate the equivalent of series capacitors.

25.12 Identify that the potential applied to capacitors in series is equal to the sum of the potentials across the individual capacitors.

25.13 For a circuit with a battery and some capacitors in parallel and some in series, simplify the circuit in steps by finding equivalent capacitors, until the charge and potential on the final equivalent capacitor can be determined, and then reverse the steps to find the charge and potential on the individual capacitors.

25-3 Capacitors in Parallel and in Series (3 of 9)

25.14 For a circuit with a battery, an open switch, and one or more uncharged capacitors, determine the amount of charge that moves through a point in the circuit when the switch is closed.

25.15 When a charged capacitor is connected in parallel to one or more uncharged capacitors, determine the charge and potential difference on each capacitor when equilibrium is reached.

25-3 Capacitors in Parallel and in Series (4 of 9)

Capacitors in Parallel

When a potential difference V is applied across several capacitors connected in parallel, that potential difference V is applied across each capacitor. The total charge q stored on the capacitors is the sum of the charges stored on all the capacitors.

$$q_1 = C_1V, \quad q_2 = C_2V, \quad \text{and} \quad q_3 = C_3V.$$

The total charge on the parallel combination of Figure. 25-8*a* is then

$$q = q_1 + q_2 + q_3 = (C_1 + C_2 + C_3)V.$$

25-3 Capacitors in Parallel and in Series (5 of 9)

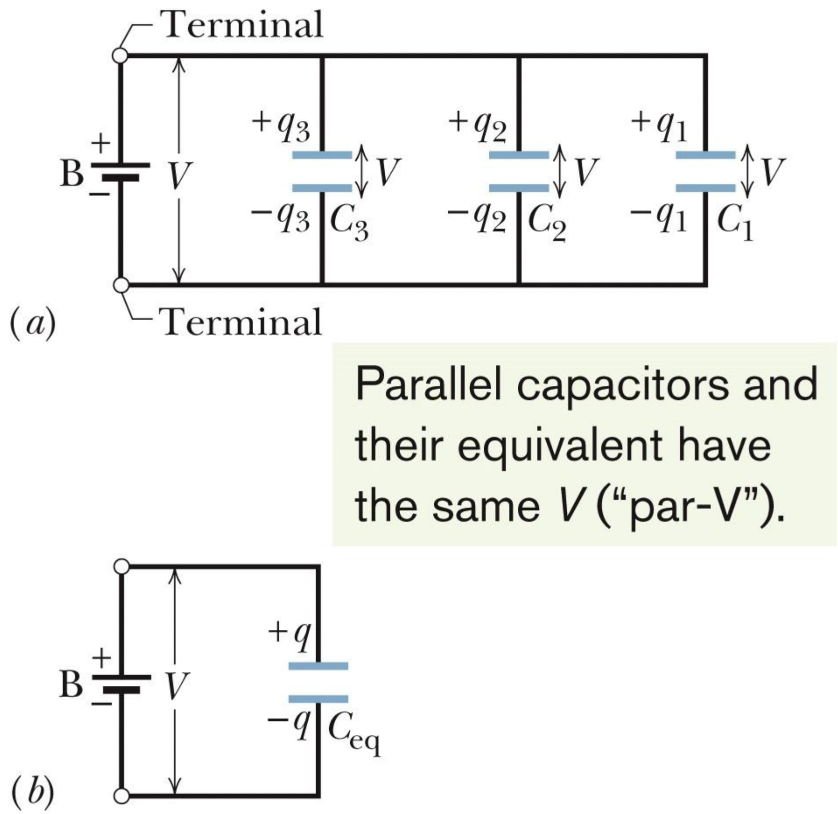
The equivalent capacitance, with the same total charge *q* and applied potential difference *V* as the combination, is then

$$C_{eq} = \frac{q}{V} = C_1 + C_2 + C_3,$$

a result that we can easily extend to any number n of capacitors, as

$$C_{eq} = \sum_{j=1}^{n} C_j \quad (n \text{ capacitors in parallel}).$$

25-3 Capacitors in Parallel and in Series (6 of 9)



Capacitors connected in parallel can be replaced with an equivalent capacitor that has the same total charge q and the same potential difference V as the actual capacitors.

25-3 Capacitors in Parallel and in Series (7 of 9)

Capacitors in Series

When a potential difference V is applied across several capacitors connected in series, the capacitors have identical charge q. The sum of the potential differences across all the capacitors is equal to the applied potential difference V.

$$V_1 = \frac{q}{C_1}, \quad V_2 = \frac{q}{C_2}, \quad \text{and} \quad V_3 = \frac{q}{C_3}.$$

The total potential difference V due to the battery is the sum

$$V = V_1 + V_2 + V_3 = q\left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}\right).$$

25-3 Capacitors in Parallel and in Series (8 of 9)

The equivalent capacitance is then

$$C_{\text{eq}} = \frac{q}{V} = \frac{1}{\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}},$$

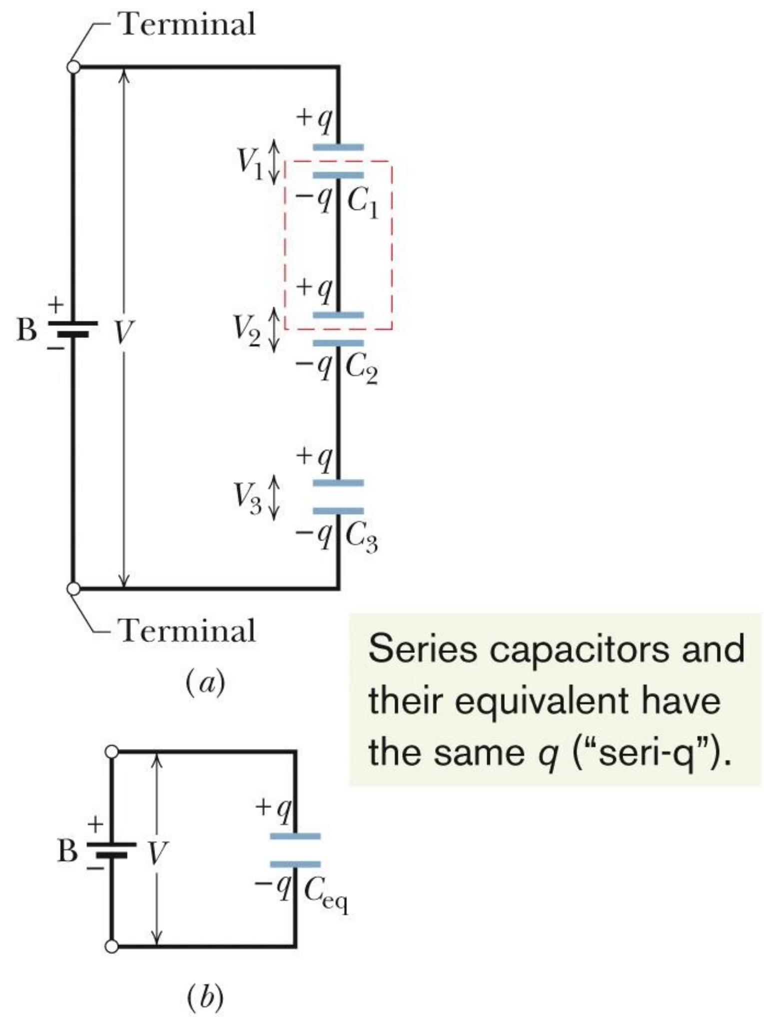
or

$$\frac{1}{C_{\text{eq}}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}.$$

$$\frac{1}{C_{\text{eq}}} = \sum_{j=1}^{n} \frac{1}{C_j} \quad (n \text{ capacitors in series}).$$

25-3 Capacitors in Parallel and in Series (9 of 9)

Capacitors that are connected in series can be replaced with an equivalent capacitor that has the same charge q and the same total potential difference V as the actual series capacitors.



Find the equivalent capacitance for the combination of capacitances

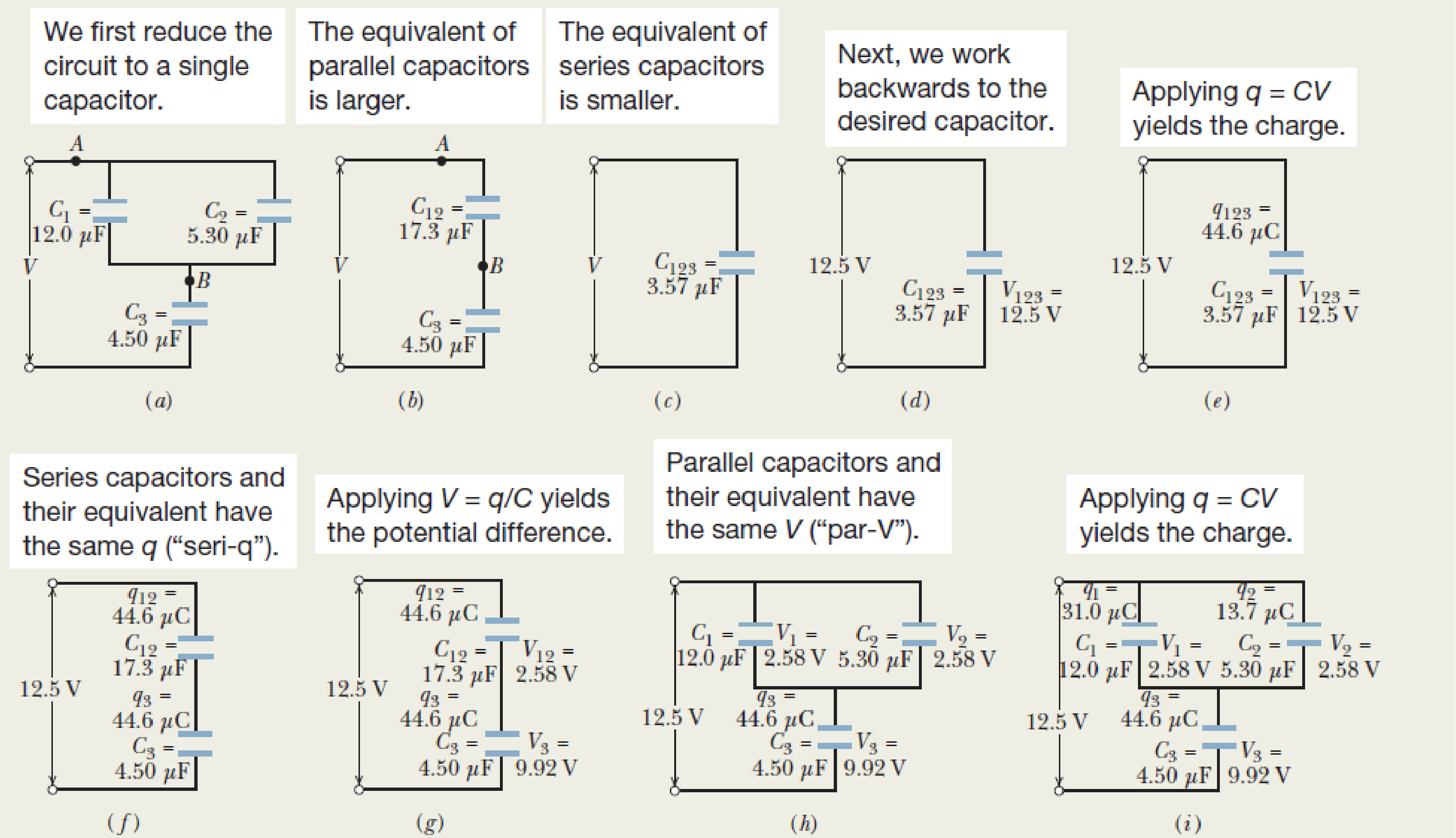


Figure 25-10 *(a)–(d)* Three capacitors are reduced to one equivalent capacitor. *(e)–(i)* Working backwards to get the charges.

bottom plates of both capacitor 1 and capacitor 2. Because there is more than one route for the shifting charge, capacitor 3 is not in series with capacitor 1 (or capacitor 2). Any time you think you might have two capacitors in series, apply this check about the shifting charge.

Are capacitor 1 and capacitor 2 in parallel? Yes. Their top plates are directly wired together and their bottom plates are directly wired together, and electric potential is applied between the top-plate pair and the bottom-plate pair. Thus, capacitor 1 and capacitor 2 are in parallel, and Eq. 25-19 tells us that their equivalent capacitance C_{12} is

$$C_{12} = C_1 + C_2 = 12.0\ \mu\text{F} + 5.30\ \mu\text{F} = 17.3\ \mu\text{F}.$$

In Fig. 25-10*b*, we have replaced capacitors 1 and 2 with their equivalent capacitor, called capacitor 12 (say "one two" and not "twelve"). (The connections at points *A* and *B* are exactly the same in Figs. 25-10*a* and *b*.)

Is capacitor 12 in series with capacitor 3? Again applying the test for series capacitances, we see that the charge that shifts from the top plate of capacitor 3 must entirely go to the bottom plate of capacitor 12. Thus, capacitor 12 and capacitor 3 are in series, and we can replace them with their equivalent C_{123} ("one two three"), as shown in Fig. 25-10*c*. From Eq. 25-20, we have

$$\frac{1}{C_{123}} = \frac{1}{C_{12}} + \frac{1}{C_3} = \frac{1}{17.3\ \mu\text{F}} + \frac{1}{4.50\ \mu\text{F}} = 0.280\ \mu\text{F}^{-1},$$

from which

$$C_{123} = \frac{1}{0.280\ \mu\text{F}^{-1}} = 3.57\ \mu\text{F}. \qquad \text{(Answer)}$$

(b) The potential difference applied to the input terminals in Fig. 25-10*a* is $V = 12.5$ V. What is the charge on C_1?

KEY IDEAS

We now need to work backwards from the equivalent capacitance to get the charge on a particular capacitor. We have two techniques for such "backwards work": (1) Seri-q: Series capacitors have the same charge as their equivalent capacitor. (2) Par-V: Parallel capacitors have the same potential difference as their equivalent capacitor.

Working backwards: To get the charge q_1 on capacitor 1, we work backwards to that capacitor, starting with the equivalent capacitor 123. Because the given potential difference V (= 12.5 V) is applied across the actual combination of three capacitors in Fig. 25-10*a*, it is also applied across C_{123} in Figs. 25-10*d* and *e*. Thus, Eq. 25-1 ($q = CV$) gives us

$$q_{123} = C_{123}V = (3.57\ \mu\text{F})(12.5\ \text{V}) = 44.6\ \mu\text{C}.$$

The series capacitors 12 and 3 in Fig. 25-10*b* each have the same charge as their equivalent capacitor 123 (Fig. 25-10*f*). Thus, capacitor 12 has charge $q_{12} = q_{123} = 44.6\ \mu$C. From Eq. 25-1 and Fig. 25-10*g*, the potential difference across capacitor 12 must be

$$V_{12} = \frac{q_{12}}{C_{12}} = \frac{44.6\ \mu\text{C}}{17.3\ \mu\text{F}} = 2.58\ \text{V}.$$

The parallel capacitors 1 and 2 each have the same potential difference as their equivalent capacitor 12 (Fig. 25-10*h*). Thus, capacitor 1 has potential difference $V_1 = V_{12} = 2.58$ V, and, from Eq. 25-1 and Fig. 25-10*i*, the charge on capacitor 1 must be

$$q_1 = C_1V_1 = (12.0\ \mu\text{F})(2.58\ \text{V}) = 31.0\ \mu\text{C}. \qquad \text{(Answer)}$$

25-4 Energy Stored in an Electric Field (1 of 3)

Learning Objectives

25.16 Explain how the work required to charge a capacitor results in the potential energy of the capacitor.

25.17 For a capacitor, apply the relationship between the potential energy U, the capacitance C, and the potential difference V.

25.18 For a capacitor, apply the relationship between the potential energy, the internal volume, and the internal energy density.

25.19 For any electric field, apply the relationship between the potential energy density u in the field and the field's magnitude E.

25.20 Explain the danger of sparks in airborne dust.

25 Summary (1 of 4)

Capacitor and Capacitance

- The capacitance of a capacitor is defined as:

$$q = CV$$ **Equation (25-1)**

Determining Capacitance

- Parallel-plate capacitor:

$$C = \frac{\varepsilon_0 A}{d}.$$ **Equation (25-9)**

- Cylindrical Capacitor:

$$C = 2\pi\varepsilon_0 \frac{L}{\ln\left(\frac{b}{a}\right)}.$$ **Equation (25-14)**

25 Summary (2 of 4)

- Spherical Capacitor:

$$C = 4\pi\varepsilon_0 \frac{ab}{b-a}. \qquad \textbf{Equation (25-17)}$$

- Isolated sphere:

$$C = 4\pi\varepsilon_0 R. \qquad \textbf{Equation (25-18)}$$

Capacitor in parallel and series

- In parallel:

$$C_{\text{eq}} = \sum_{j=1}^{n} C_j \qquad \textbf{Equation (25-19)}$$